

Quick Start Manual: OpenBCI Equipment Setup

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This document will guide you through the initial setup of the OpenBCI hardware and software equipment development environment as well as connecting the OpenBCI GUI application software to the Ultracortex Mark IV helmet.



Figure 1 OpenBCI biosensing EEG headset.

Prerequisites:

Before beginning, ensure you carry the necessary items ready:

→ Software

1. OpenBCI GUI.
2. Processing Java software.

→ Hardware

1. Ultracortex Mark IV 3D-printed EEG helmet.
2. 8 Spikey Units.
3. 2 Flat Units.
4. 7 Comfort Units.
5. 3 Ribbon Cables.
6. 2 Ear Clips.
7. OpenBCI USB Dongle.
8. Cyton Board.
9. Laptop to pair the 8-electrode connections between the EEG helmet & software.

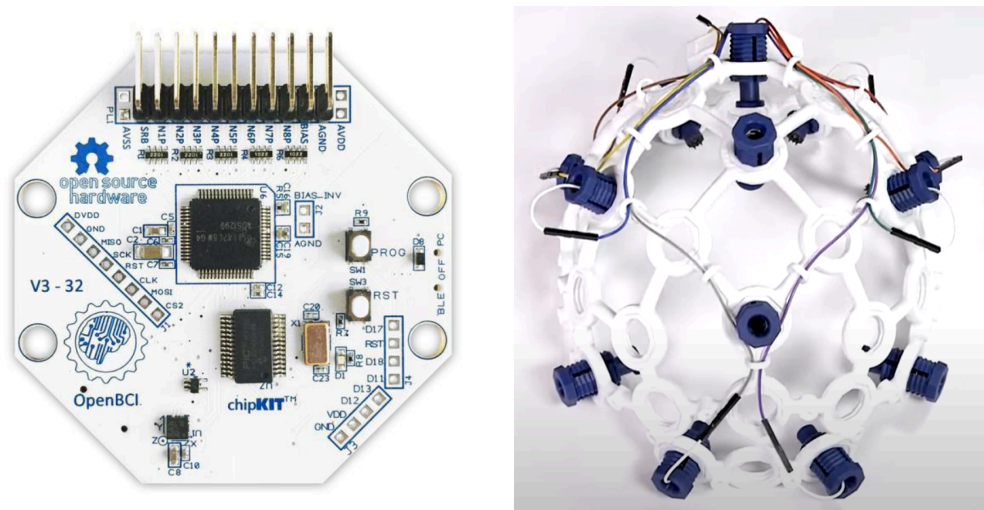


Figure 2 OpenBCI Cyton board (left) and OpenBCI EEG assembled headset (right).

NOTE: If the helmet is **NOT** completely **assembled** it should be needed:

1. 1 Frame.
2. 35 Inserts.
3. 30 Wired Clips.
4. 1 Board Mount.
5. 1 Board Cover.
6. Superglue.
7. Cleaning tool.
8. 2 #4 Screws.
9. OpenBCI Board: Could be Cyton, Daisy, Cyton & Daisy or Ganglion.

Use the cleaning tool to clear the headset before flushing the insert components. The front and back frames must then be glued together afterwards. Nodes should be glued and pasted together onto a flat surface. Once the frame is glued, the inserts can be attached to it. Then, using a screwdriver, mount the board to the back side of the frame. Further information is detailed at the following **Ultracortex Mark IV** section.

Ultracortex Mark IV

The Ultracortex is an open-source, 3D-printable headset designed to work with OpenBCI EEG biosensing Boards. It is assembled by 3D printed parts as well as non-printed pieces:

3D-printed Parts

- FRAME (head circumference front and back).

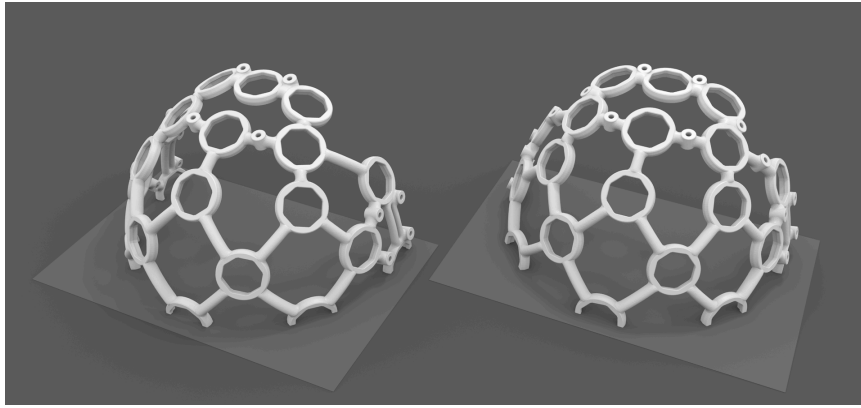


Figure 3 Headset frames.

- BOARD MOUNT

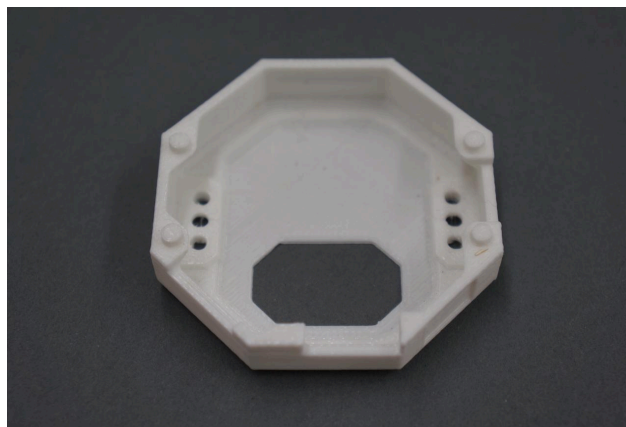


Figure 4 Mount piece for Cyton board.

- BOARD COVER

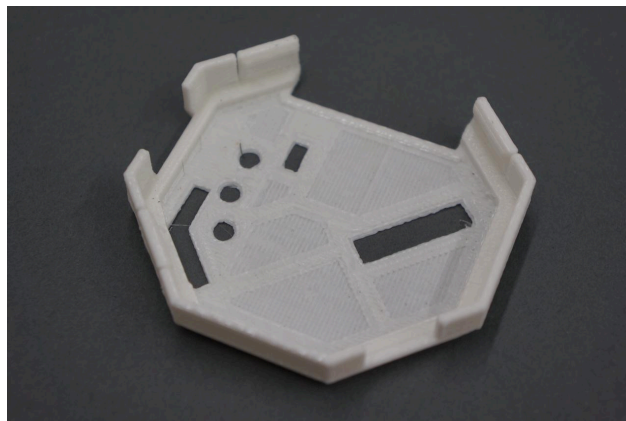


Figure 5 Cyton board cover.

Non-printed Parts

- No.6 screws for brittle plastic.



Figure 6 Screw required type.

- Cables

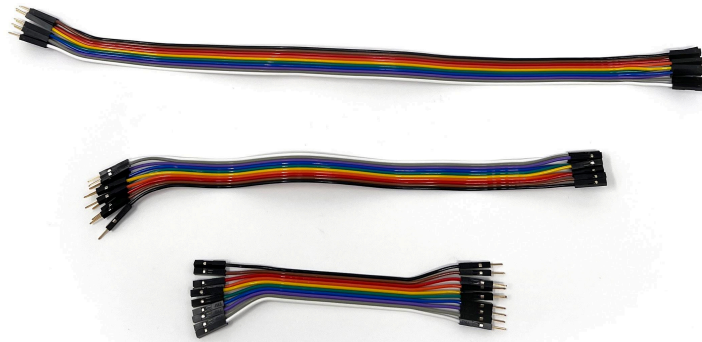


Figure 7 Provided headset cables.

- Spikey units: these electrodes are installed in Ultracortex nodes with hair.



Figure 8 Spikey units electrodes.

- Flat units: these electrodes are installed in Ultracortex nodes without hair (forehead, for instance).



Figure 9 Flat units electrodes.

- Comfort units: are used for distributing headset weight equally.



Figure 10 Comfort units electrodes.

- Ear Clips electrodes.



Figure 11 Earclips electrodes.

- An OpenBCI Cyton 8-channel Board.

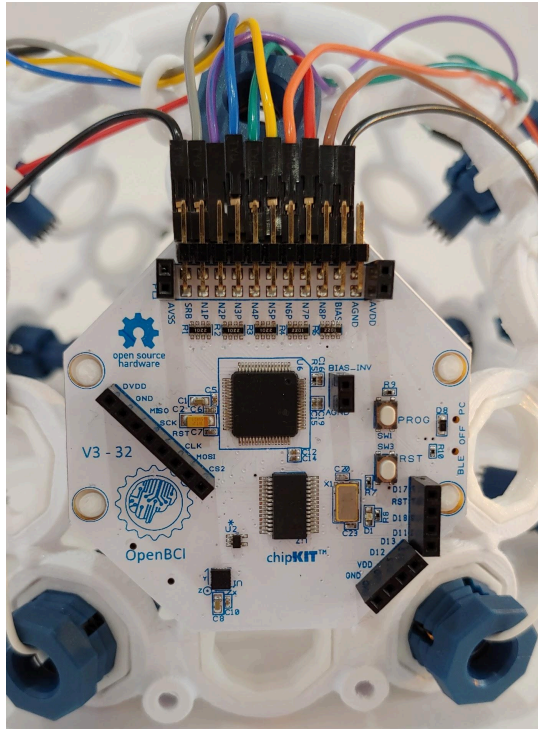


Figure 12 Cyton wires connected to board visualization.

- Lithium Ion Rechargeable Battery Pack (~500mAh)

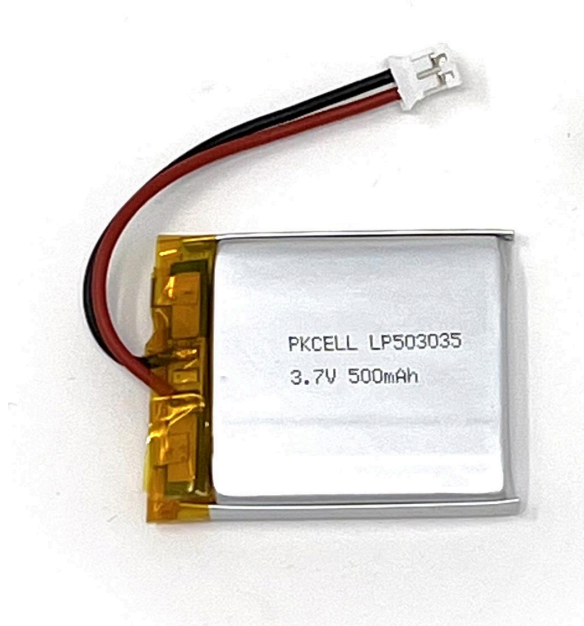


Figure 13 Lithium Cyton required battery.

- A charger for the battery pack.
 - Velcro One Wrap.
-

Electrode Assembly

- Comfort Units are set according to the 10-20 International System: **F7, F8, Fz, Cz** and **Pz**.
- Flat Electrodes are placed at **Fp1** and **Fp2** node locations.
- Spikey Electrodes are placed on positions: **C3, C4, P7, P8, O1** and **O2**.

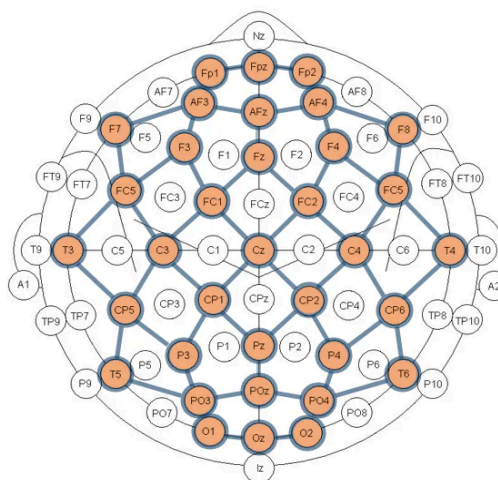
For the **8-electrode configuration**:

Long cables are firstly peeled off. Grey, **White** and **Purple** are separated and **White** one is put aside. Grey wire is attached then to the **Fp2** electrode and **Purple** to **Fp1**.

From the Medium Cables it is stripped off the **Green** and **Blue** wires. **Blue** wire goes into **C3** and **Green** into **C4**. Then, from the medium cables again, **Orange** goes into **P7** electrode and **Yellow** into **P8**. The same process is repeated with **Red** and **Brown** cables connected to **O2** and **O1** respectively. By using the wired clips the cables are run back to the board.

Earclips can also be secured by using the wire clips on both sides respectively. Check the **Ultracortex Mark IV** section for additional information.

Ultracortex Mark IV Node Locations (35 total)



Based on the internationally accepted **10-20 System**
for electrode placement in the context of EEG research

Figure 14 10-20 International electrode placement system.

Electrode location overview

The Ultracortex node locations are based on the 10-20 International System, which is the one accepted and considered as standard for electrode placement in the context of EEG.

The Ultracortex Mark IV headset should be placed on the head so that the back center node is at the same distance above the inion (bump on the back of the skull) and the front center node is above the bridge of the nose. While placing the Ultracortex headset on the head, the springs should be adjusted to the convenient shape and size of the user wearing it.

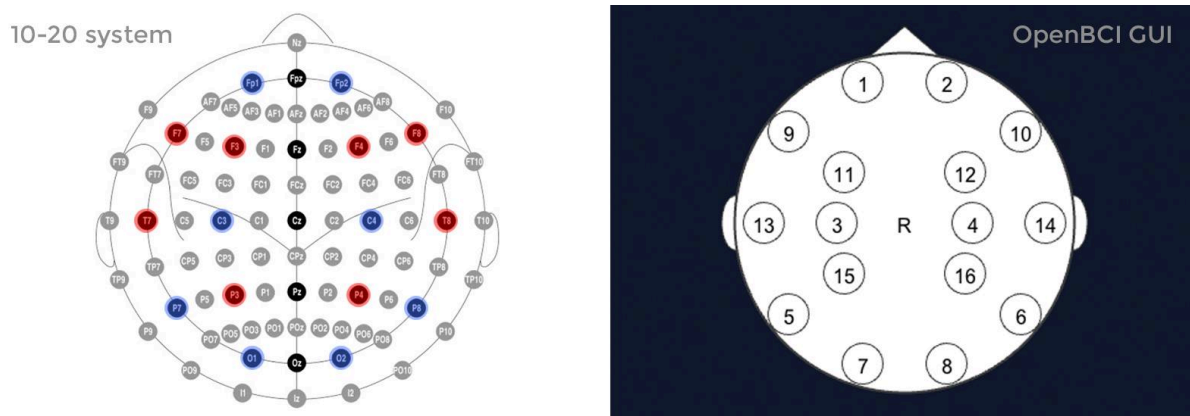


Figure 15 10-20 Electrode International System and GUI electrode locations visualization.

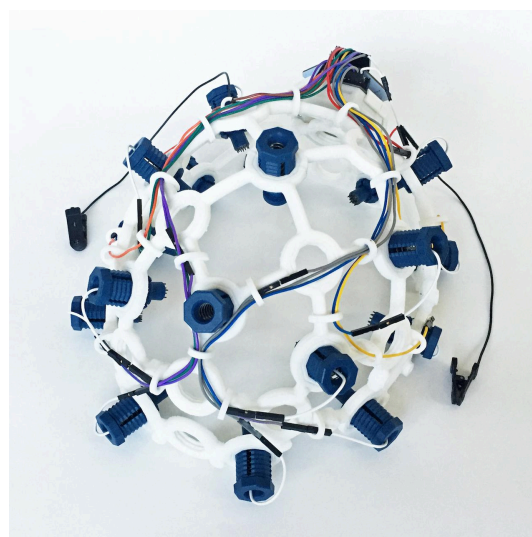


Figure 16 Upper view of OpenBCI Ultracortex Mark IV headset.

Ear Clips

To use the Mark IV headset, there are needed two ear clip electrodes, which come with the headset kit. These ones serve as the reference and bias (ground with common-mode noise rejection) for the EEG system. The electrodes (FP1 through O2) can be connected to any N1P, N2P, etc, channel. The ear clips must always be connected to the bottom SRB pin and either of the BIAS pins.

NOTE: To connect the wires to the electrodes, plug the male end into the header cable on the NODES as shown on the image above.

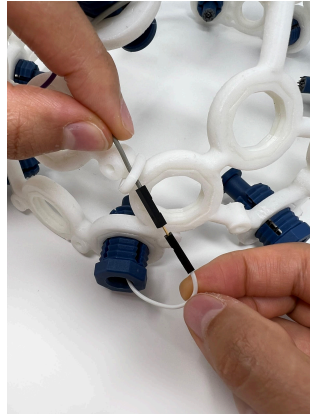


Figure 17 Wire-electrode connection management (male-header link).

OpenBCI Cyton board

1. Cyton Board Specifications

- Power: 3-6V DC Battery ONLY
- PIC32MX250F128B Microcontroller with chipKIT UDB32-MX2-DIP bootloader
- ADS1299 Analog Front End
- LIS3DH 3-axis Accelerometer
- RFduino **BLE** radio
- Micro SD card slot
- Voltage Regulation: 3.3V, +2.5V, -2.5V.
- Board Dimensions: 2.41" x 2.41" (octagon has 1" edges).
- Mount holes: 1/16" ID, 0.8" x 2.166" centre-to-centre.
- Input voltage range: +2.5 to -2.5V.
- Signal-Noise-Ratio: 121 dB.
- Voltage resolution = 0.298 microvolt/bit (5V/16.777.216).
- Resolution = Voltage level / 2^n where n is the bit-level of ADC.

2. Slide Switch

The slide switch powers the board. It has three positions:

- BLE: activates a pull-up on RFduino pin 4.
- OFF: disconnects the battery input.
- PC: does NOT activate a pull-up on RFduino pin 4.

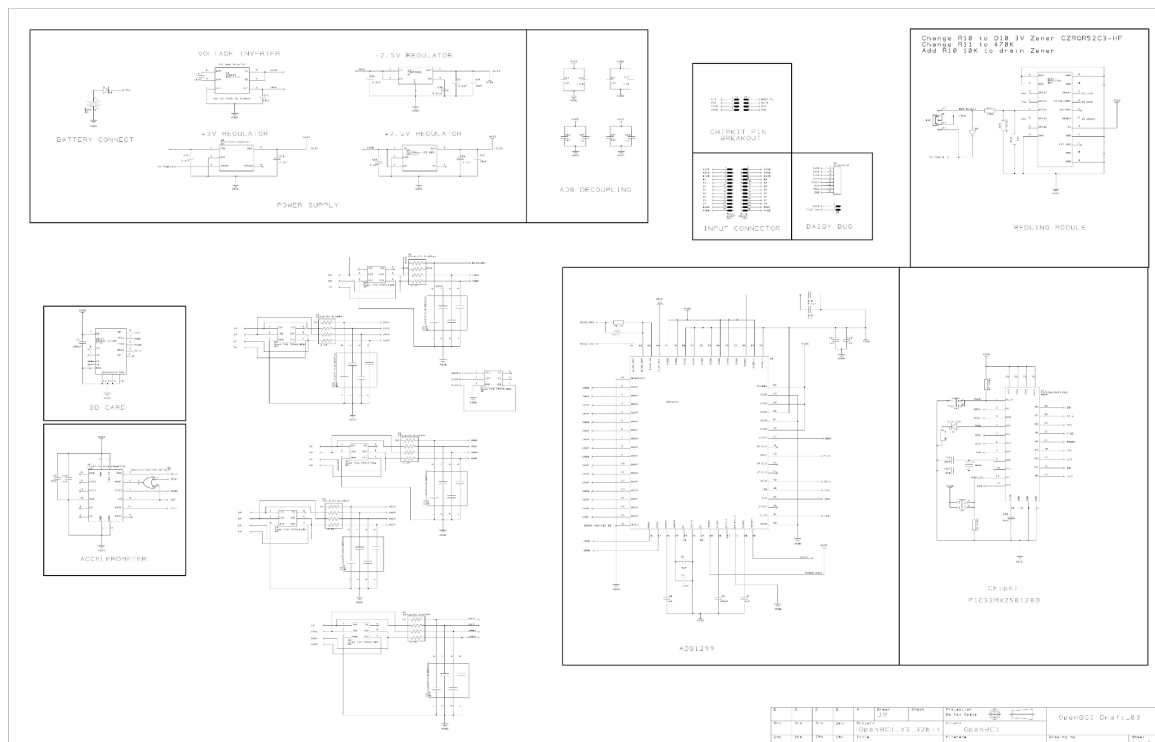


Figure 18 OpenBCI Cyton board schematic circuit.

Connection between the Mark IV EEG headset and the Cyton Board

The cables must be connected to each Cyton bottom pin such that it follows the next provided table located at the next page.

Afterwards, connect the **REFERENCE** electrode to the **Y** cable. If the **Daisy** Board is also plugged in, connect the female ends of the **Y splitter cable** to the bottom **SRB pins** on both Cyton and Daisy boards. Then connect the male pin of the Y cable to an ear clip electrode.

Once everything is assembled, place the headset on your head correctly, ensuring that the center node lines up with your nose. While putting on the helmet, spikey units must be adjusted to gather a clean signal. Comfort nodes should be adjusted until they touch your scalp. Then ear clips must be connected to the earlobes.

Electrode Node	Cyton Pin
O2	N8P
O1	N7P
P8	N6P
P7	N5P
C4	N4P
C3	N3P
Fp2	N2P
Fp1	N1P

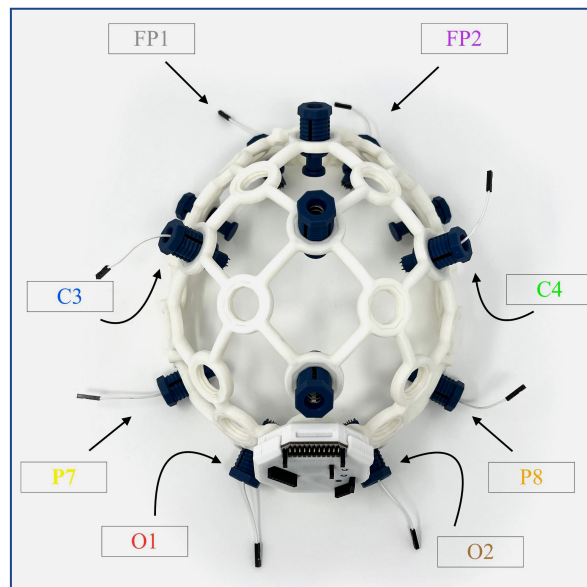


Figure 19 Ultracortex Mark IV electrode's placement.

OpenBCI USB Dongle

The OpenBCI USB Dongle is primarily used to connect your computer to the Cyton board. It has an integrated RFDuino that communicates with the RFDuino on the Cyton board. The dongle establishes a serial connection with the FTDI chip on your computer's board. This serial port is labelled as COM* in Windows. You will connect to this serial port from the OpenBCI GUI or whatever other software you intend to use to interface with your Cyton board. The dongle is plugged in on the laptop in use. The LED must be seen ON and with a blue color when this occurs.

1. Dongle Specifications

- Power via USB connector **ONLY**
- RFduino **BLE** radio module
- FTDI USB-Serial IC (FT231XQ-R)
- Resettable fuse

2. Breakout Pins

The RFduino pins are broken out in the same order and layout as the RFduino radio and shields. This makes the OpenBCI USB Dongle compatible with the RFduino boards. The TXD (red) and RXD (green) LEDs are connected to outputs from the FTDI chip. The blue LED is connected to RFduino GPIO2.

3. Slide Switch

The slide switch on the Dongle has two positions. When the switch is on the GPIO6 side, the FTDI DTR pin is routed to the RFduino pin 6, allowing data to be passed to and from the Cyton board. This configuration is ‘usual’ mode, and also allows for programming the Cyton board via wireless connection. When the switch is on the RESET side, the FTDI DTR pin is routed to the RFduino RESET pin. This mode enables the RFduino on the Dongle to be reprogrammed.

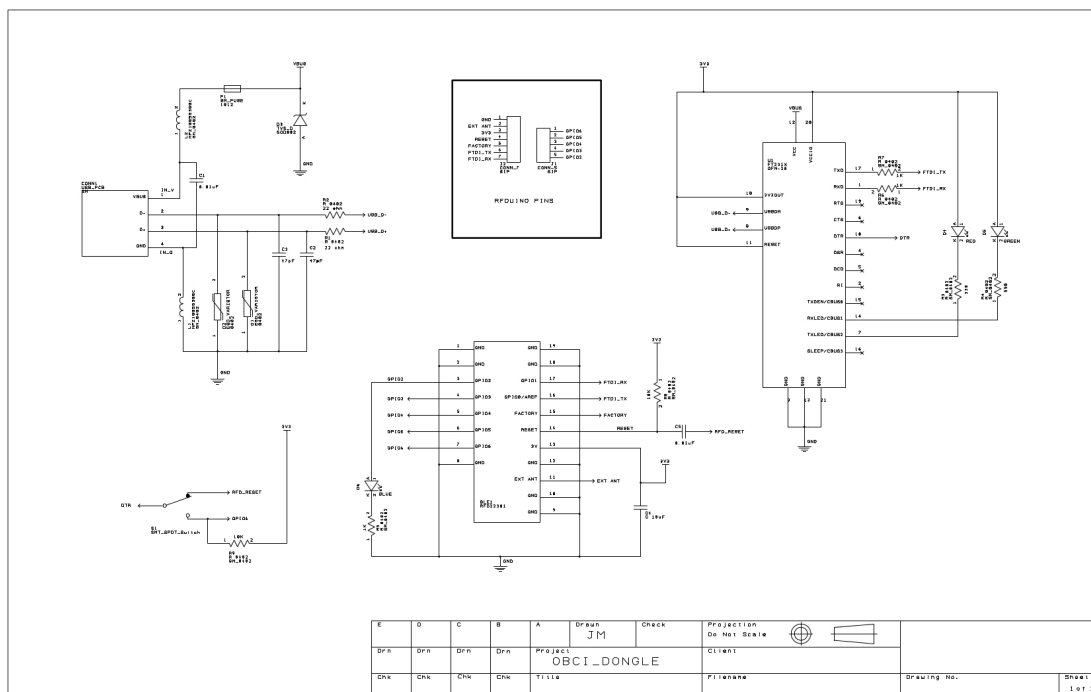


Figure 20 OpenBCI Dongle schematic circuit.

Connection between the OpenBCI USB Dongle & the Cyton board

Firstly, ensure that your OpenBCI USB Dongle is plugged in facing upwards. A blue LED will light up and stay on as well as a red LED flash will be observed.

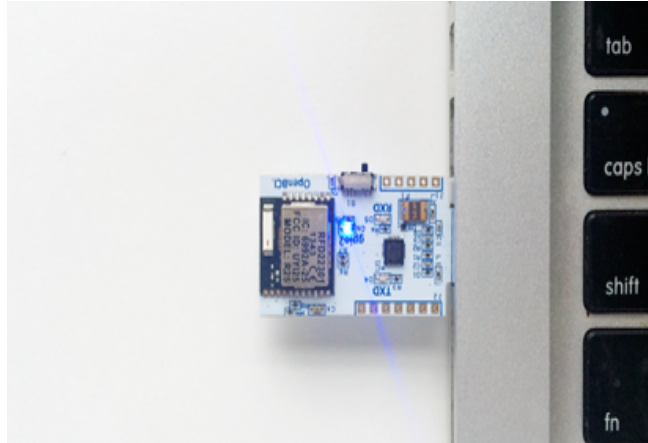


Figure 21 USB Dongle connection visualization.

NOTE: Ensured that the USB Dongle is switched to **GPIO 6** and NOT to RESET mode. To achieve this, the switch should be set closer to the computer as it shows the upwards figure.

The next step is to switch ON the Cyton board to **PC mode** (not OFF or BLE configurations).

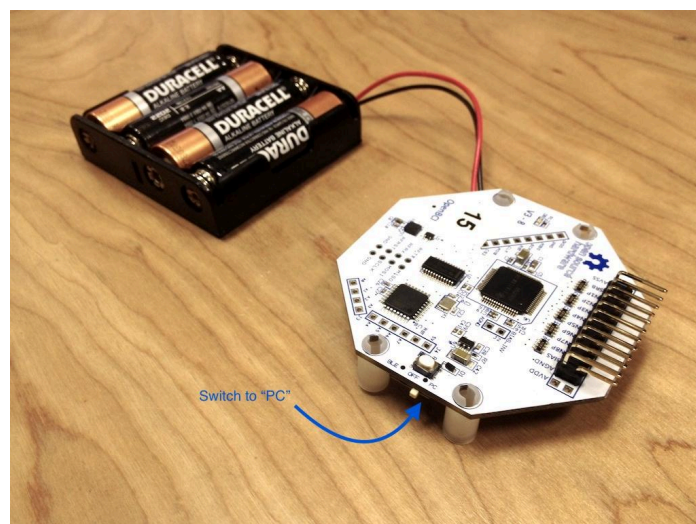


Figure 22 Switch PC Cyton board configuration.

When moving the small switch on the right-hand side of the board from "OFF" to "PC", a blue LED should turn on. If not, press the reset (RST) button to the left of the switch. If the LED still does not turn on, check that the board has a full battery.

Dongle must be plugged in before the Cyton board is turned on. If the data stream appears as broken, you may need to unplug the USB Dongle and power down the Cyton board. Make

sure to plug the USB Dongle in first and then power up your board afterwards to register data accurately.

Connection between the Cyton board and the GUI application

In order to connect the Cyton board with the GUI it needs to be firstly selected the Data Source type. It is advisable to choose LIVE (from Cyton) displayed in the first section of the System Control Panel to see the real-time EEG signal acquisition.

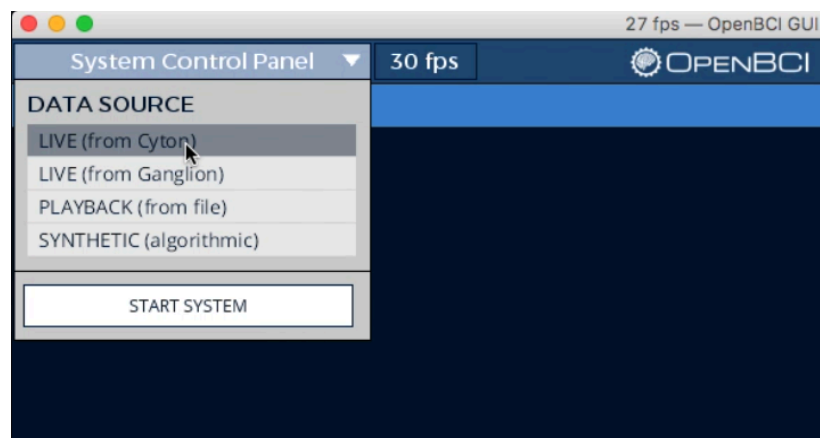


Figure 23 System Control Panel visualization.

Next, it is usually selected the Serial Transfer Protocol that is achieved thanks to the use of the USB Dongle.

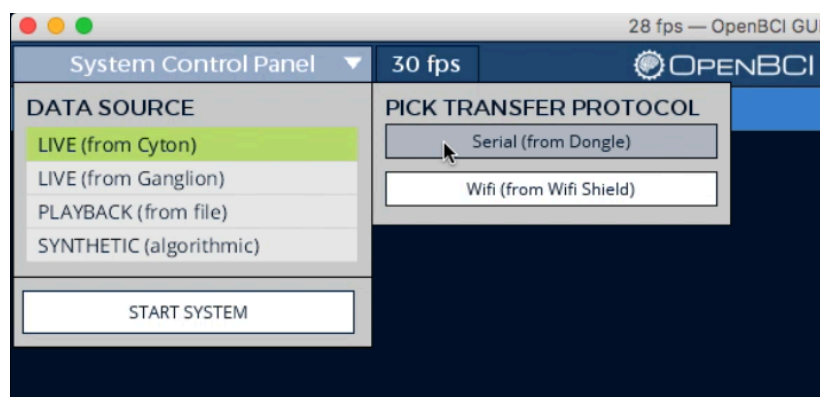


Figure 24 Transfer Protocol Panel configuration.

Once the communication transfer protocol is chosen, the available port linked to the Dongle must be pressed. AUTO-CONNECT button can be selected as seen in the image above and then AUTO-SCAN to solve in an easy manner Cyton connection troubleshooting issues.

NOTE: Otherwise these configurations must be taken if the process of finding the USB Dongle port name is achieved manually:

→ If you're using a **Mac** or **Linux**, its name will be in the following format: **/dev/tty***

→ If you're using **Windows**, it will appear as: **COM#**

The Dongle port name will likely be at the top of the list. If not, you may need to ensure that the Dongle is plugged in as well as switched to GPIO 6 (not RESET). Another option consists of clicking the REFRESH LIST button in the SERIAL/COM PORT section of the sub-panel.

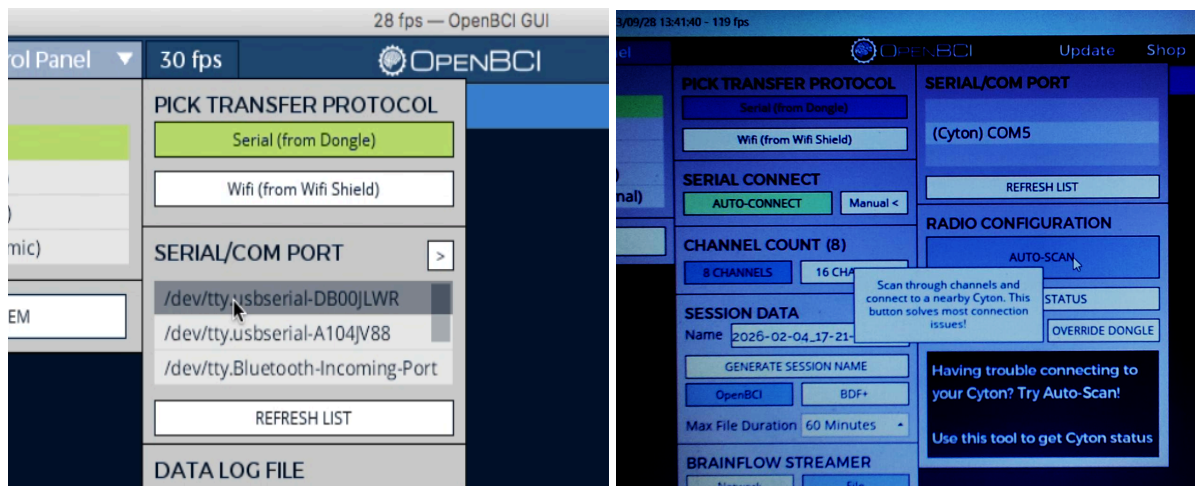


Figure 25 Searching for port availability synchronisation with a Mac device (left) and with a Windows laptop (right).

The next step is based on selecting the number of channels. By default, the CHANNEL COUNT setting is assumed as 8 channels. If it is used the Daisy Module adhered with the Cyton board it will comprise in total 16 channels, therefore this option must be changed in this scenario.

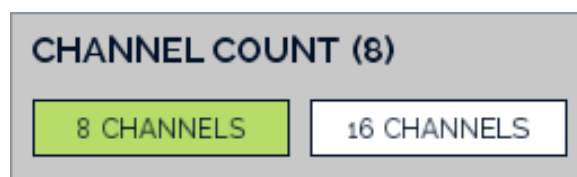


Figure 26 Channel Count selection.

At this stage the system can be perfectly started. The START SESSION button must be pressed therefore to establish a connection with the Cyton board. This process tends to take a few seconds and a message is usually displayed such as: “Attempting to establish a connection with your OpenBCI Board ...”

Furthermore, as it was not mentioned before, if there is not any additional SD card for storage each recording session it must be pressed “Do not write to SD” button. Then it should be pressed the START SESSION/SYSTEM button. Once that one is clicked, a new session configuration appears.

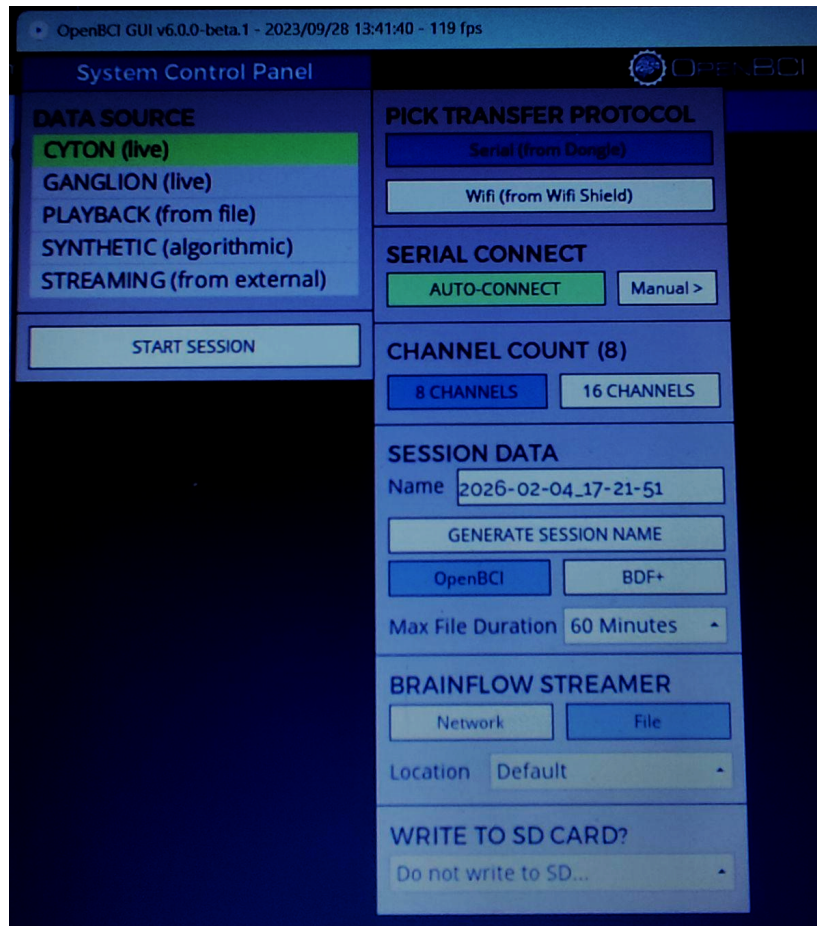


Figure 27 System Control Panel configuration.

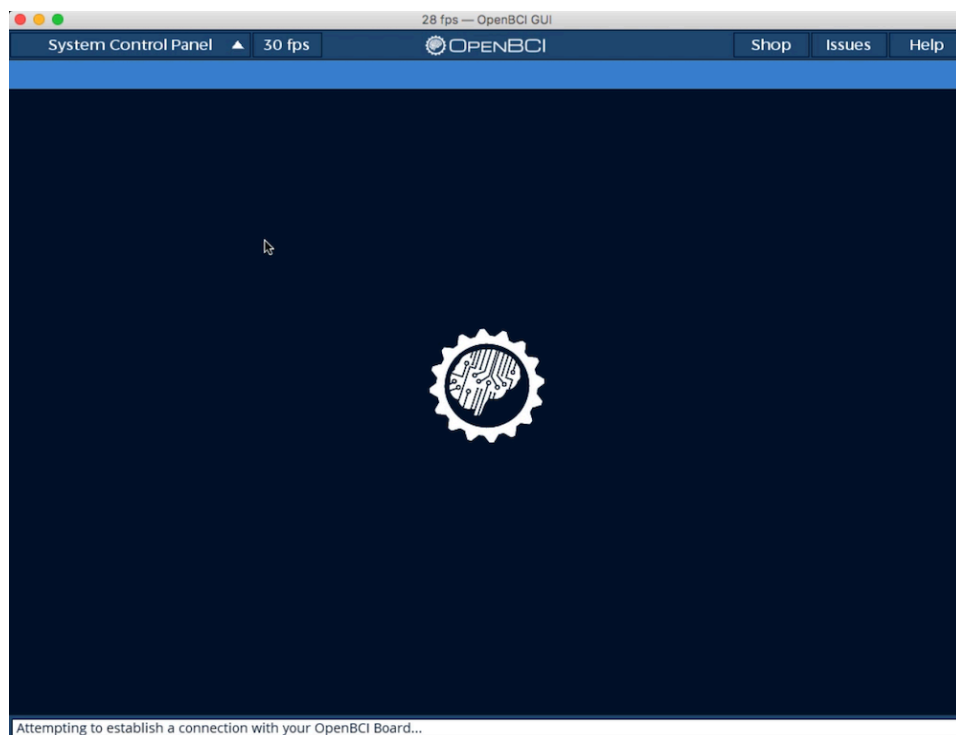


Figure 28 Attempting to enable communication between the laptop and OpenBCI Cyton board.

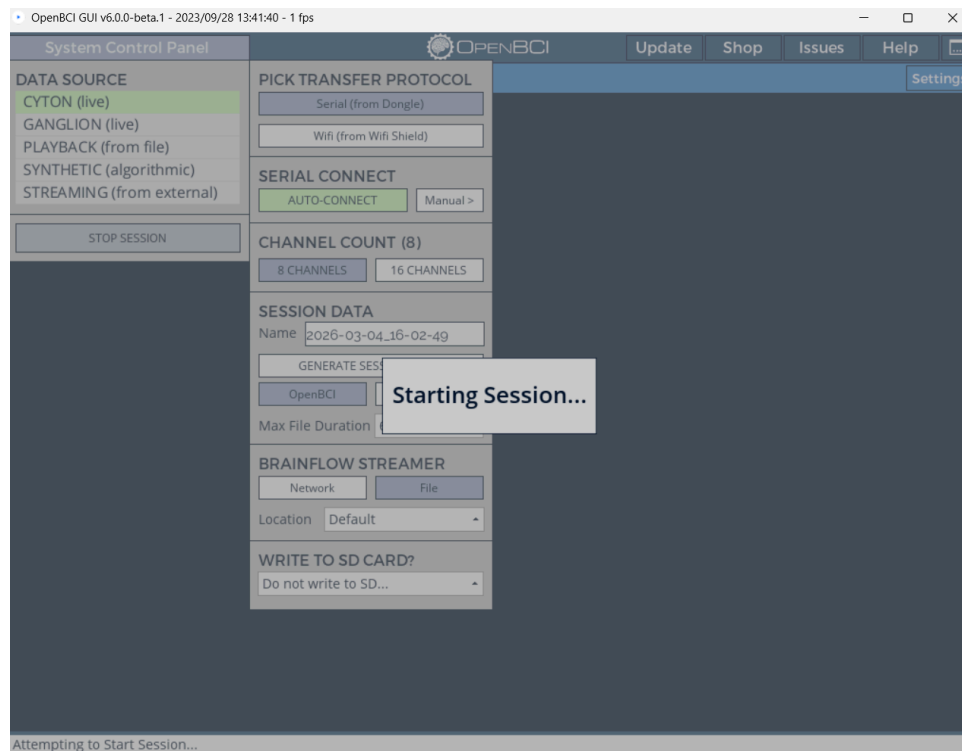


Figure 29 Beginning new session visualization.

To start recording EEG signals press Start Data Stream, which is located in the upper left hand corner:

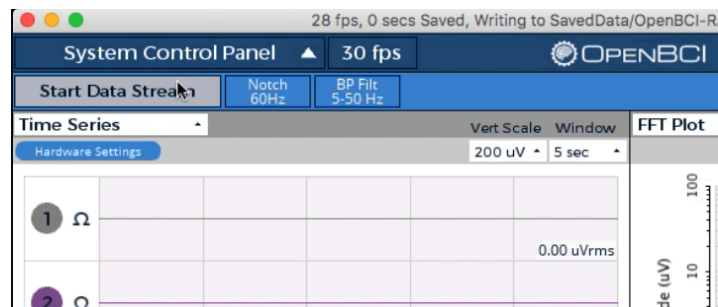


Figure 30 GUI session displayed.

As it has been connected to the Cyton successfully, the GUI displays Time Series, FFT Plot among others real-time running widgets that are explained further on the Bachelor Thesis documentation.

To begin a new recording, the Start Data Stream button must be pressed. When it is desired to end the signal registration process, it will be seen at the top left corner a button that must be pressed named Stop Data Stream.

NOTE: By default, the GUI stores the recorded raw data from each initialised session in [\[USER\]/Documents/OpenBCI_GUI](#) laptop user pathway address and names each session with an autogenerated timestamp by default, indicating the day and time of the stream session. As seen in Figure 31, the information is saved as .txt and as a .CSV file.

Name	Date modified	Type	Size
BrainFlow-RAW_2023-05-03_15-43-03_0	5/3/2023 3:48 PM	Comma Separated V...	7,319 KB
BrainFlow-RAW_2023-05-03_15-43-03_1	5/3/2023 3:58 PM	Comma Separated V...	42,136 KB
BrainFlow-RAW_2023-05-03_15-43-03_2	5/3/2023 3:59 PM	Comma Separated V...	825 KB
OpenBCI-RAW-2023-05-03_15-45-07	5/3/2023 3:46 PM	Text Document	8,053 KB
OpenBCI-RAW-2023-05-03_15-48-02	5/3/2023 3:58 PM	Text Document	46,187 KB
OpenBCI-RAW-2023-05-03_15-58-49	5/3/2023 3:59 PM	Text Document	903 KB

Figure 31 Files saved after a data session recording.

The generated raw data is managed by a Python build tool named BrainFlow (see Figure 32), which is based on a collection of code libraries that enable the signal processing information between the Cyton board and the computer being used. Therefore, there is no need for knowledge regarding the data format board communication in order to manage the provided OpenBCI GUI equipment.

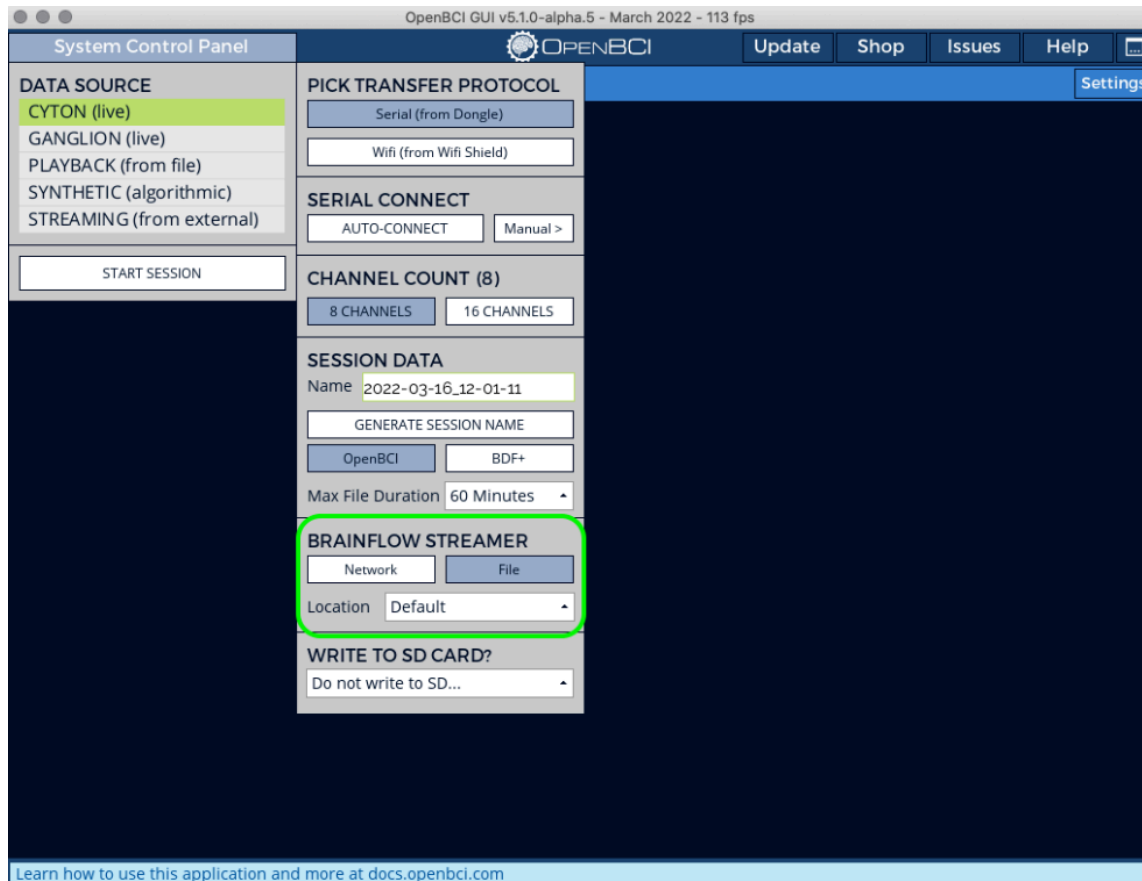


Figure 32 Enabling BrainFlow data streaming.

As only 8 channels are available, observing the Figures 34 (MAC) and 38 (Windows) it is seen the GUI appearance, resulting from the real-time stream connection done within the Ultracortex Mark IV headset with the OpenBCI Cyton board.

Several widget visualization options can be managed by the use of this application. All of them are displayed by clicking for example on the FFT Plot drop-down arrow menu placed at the top right corner:

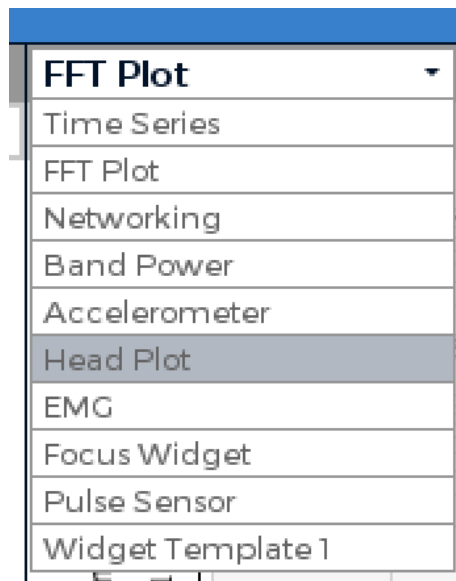


Figure 33 GUI Widget's availability.

Additionally, to visualize a map of the signal strength at each electrode, it must be selected the Head Plot Widget (see Figure 34).

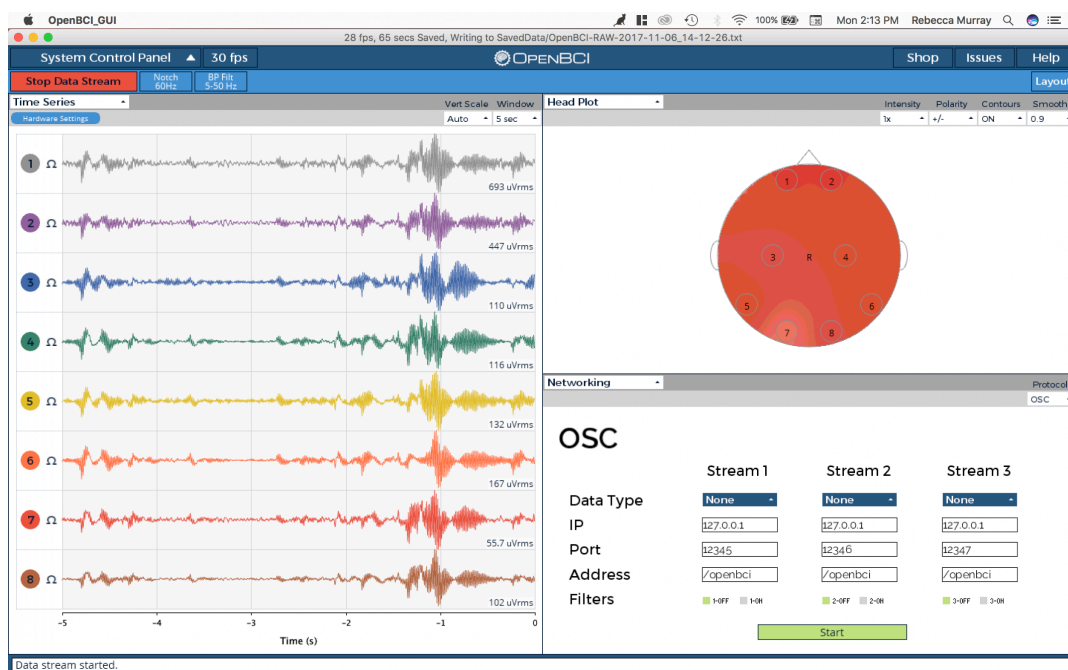


Figure 34 Head Plot, Time Series and Network Protocol widgets.

As a brief summary, the main available GUI visualisations are:

1. **Time Series:** displays the real-time electrophysiological biosensing data. Each graph measures the absolute amplitude of the signal in voltages with respect to each electrode and in μVrms units.
2. **FFT:** among the x-axis it displays the frequency domain and at the y-axis shows each frequency's respective amplitudes in μV . Furthermore, each color in the FFT Plot matches the channels represented in the Time Series widget.
3. **Band Power:** shows the relative voltages of the brain waves. Each type of brain wave represents a subset of frequencies, which represent different states of activity: Gamma waves ($> 32 \text{ Hz}$) associated with alertness and focus perception; Beta waves ($13\text{-}32 \text{ Hz}$) linked with active thinking; Alpha ($8\text{-}13 \text{ Hz}$) related with relaxation states such as eyes close conditions; Theta ($4\text{-}8 \text{ Hz}$) linked with deep meditation scenarios and Delta ($0.5\text{-}4 \text{ Hz}$) related to sleep stages.

NOTE: the filtering range should be adjusted at a minimum to a 0.5 Hz value in order to capture correctly the Delta frequency band.



Figure 35 Complete GUI main widgets visualization.

4. **Filter configuration:** by pressing the Filters button placed close to the Hardware Settings panel at the top left side several built-in filters can be chosen. It can be selected the frequency range, type (Butterworth, Chebyshev, Bessel) and order of the filters. The notch filter option is enabled to reduce the AC noise by filtering out 50 Hz or 60 Hz noise. There is also a dropdown menu to choose to apply a bandpass or bandstop filter to the data. A new filter configuration can be edited as well as saved instead of the use of the default values.

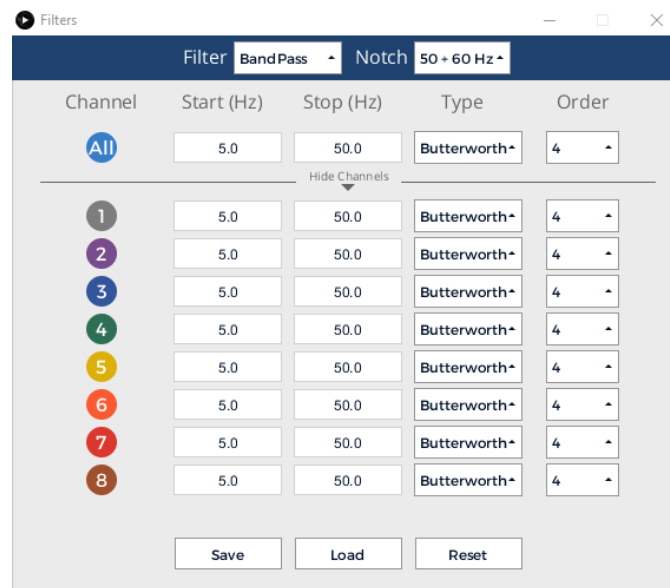


Figure 36 Filter configuration.

5. **Impedance check:** this widget is a tool for evaluating the electrode quality and contact before data acquisition. Once the Check all channels button is pressed it starts the impedance checking on an individual channel. A red impedance value means the electrode needs to be adjusted better. Also during live acquisition some channels may show a Rail warning message. This one indicates that there is a poor signal collected due to exceeding the maximum signal amplitude's value. Impedance testing should be therefore applied to ensure that a good contact with the scalp is achieved. Also, during the impedance test it is not able to stream data. It is hence recommendable to use it before starting a recording session.

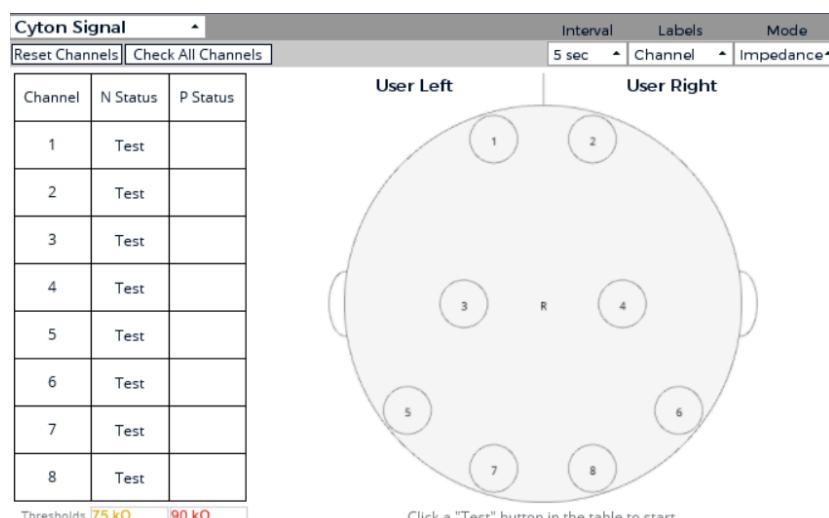


Figure 37 Impedance check visualization.



Figure 38 Live impedance testing visualization.